



Đinh Đức Hiệp

CONTACT

Phone: +84 977963005 **Address:** Ha Dong, Hanoi
Email: hiepdinh.dd@gmail.com

ABOUT ME

Undergraduate at Hanoi University of Science and Technology and an AI Engineer building production LLM systems. I work with Python/PyTorch across machine learning and deep learning: I built a CNN-GRU model for bearing fault diagnosis reaching over 99% accuracy, and RAG-based LLM systems in production, including an automated SEO content pipeline and an AI product-comparison assistant. My current focus is agentic AI and computer vision.

I learn by reading technical documentation, running my own experiments, and shipping working systems — and I want to bring that hands-on approach to my campus, building a community where students move from using AI tools to building with them.

EDUCATION

PFIEV – Industrial Informatics and Automation
Hanoi University of Science and Technology | 2023 – Present

SKILLS

- Have fundamental knowledge of AI/LLM, RAG, Prompt Engineering, and vector search; practically applied through two projects: an AI system for generating and publishing SEO content and the AI Product Comparison Advisor
- Basic programming: Python, C/C++; basic knowledge of HTML, CSS, and MySQL
- Have a foundation in Machine Learning/Deep Learning with PyTorch and signal processing with FFT/STFT through a practical CNN-GRU project
- Have experience in building backends and APIs using FastAPI, SQLite/PostgreSQL, ChromaDB as a vector database, and WebSocket
- Able to use AI agents to support coding, debugging, reading documents, and quickly creating demos
- Able to design system architecture, including embedded systems and software, with system thinking and the ability to quickly grasp new technologies

MACHINE LEARNING & LLM PROJECTS

Design of a Device Using a Deep Learning Model for AC Motor Bearing Fault Warning

- A device using a Deep Learning model combined with vibration signal analysis to diagnose bearing faults in AC motors
- Current progress: completed the CNN-GRU Deep Learning model based on the existing CWRU, MFPT, and Paderborn datasets
- Model results after training:
 - Merged data from three datasets: accuracy of 99.6%
 - CWRU data: accuracy of 99.79%
 - MFPT data: accuracy of 100%
 - Paderborn data: accuracy of 99.85%
- Technologies used: Python, PyTorch, CNN-GRU, CWRU dataset, MFPT dataset, Paderborn dataset

AI System for Automatically Generating and Publishing SEO Content Using LLM and RAG

Link github: <https://github.com/Dinhduchiep30092005/chatbot-LLM-RAG>

- An automated system using an LLM, GPT-4o, combined with RAG techniques to research keywords and generate SEO-standard articles
 - Built a “knowledge memory” for AI by reading and extracting information from available PDF/DOCX documents, helping the generated content become more accurate and in-depth, following E-E-A-T standards
 - Integrated SerpAPI to analyze SERP and the Google Search Console API to monitor keyword performance and suggest optimizing titles and meta descriptions when CTR is low
 - Output:
 - Complete SEO articles by topic cluster, including one pillar article and five to ten internally linked cluster articles, together with data tables, FAQ schema, and technical comparisons
 - Automatically generates illustrative images for each article
 - Automatically publishes to WordPress/Shopify in multiple languages, including categories, tags, and featured images
 - Technologies used: Python, OpenAI API with GPT-4o, SerpAPI, PostgreSQL/SQLAlchemy, Flask/Gunicorn, sentence-transformers, pdfplumber/python-docx, Pillow, Google Search Console API, WordPress/Shopify API
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AI Product Comparison Advisor

Competition project: VietNam AI Innovation Challenge 2026

Demo link: <https://github.com/Dinhduchiep30092005/AI-Product-Comparison-Advisor>

- Built an AI assistant for consulting on electrical appliance products in natural Vietnamese, including typing without diacritics and abbreviations; automatically asks follow-up questions to clarify needs (budget, intended use, etc.) instead of relying on sales staff experience as in the traditional approach
- Designed the processing pipeline: Slot-filling to extract needs → Vector search (ChromaDB, rerank) → Domain Rule Engine (manually defined rules for 19 core product categories, covering approximately 83% of the catalog) → LLM generates responses verified against real data, preventing AI from “making up” price/stock information
- Integrated real-time alerts via WebSocket, proactively notifying customers when products they are interested in have price reductions or are back in stock; stored each customer’s consultation history across multiple chat sessions (cross-session memory by customer_id, stored in the database)
- Output:
 - Always recommends the Top 3 most suitable products with a visual comparison table and clearly explains the reasons for the selection
 - Provides real-time updates on events such as price changes, discounts, products being back in stock, etc
 - Administration dashboard: manages products/inventory/prices in real time, manages policy documents, sends broadcast promotions, and records audit logs of all admin operations and conversations
- Technologies used: Python, FastAPI, ChromaDB (multiple vector stores for RAG), DeepSeek-V4-Flash LLM (FPT AI Marketplace, fallback gpt-oss-120b), Vietnamese_Embedding, bge-reranker-v2-m3, React 19/TypeScript/Tailwind CSS, WebSocket, SQLite, Docker